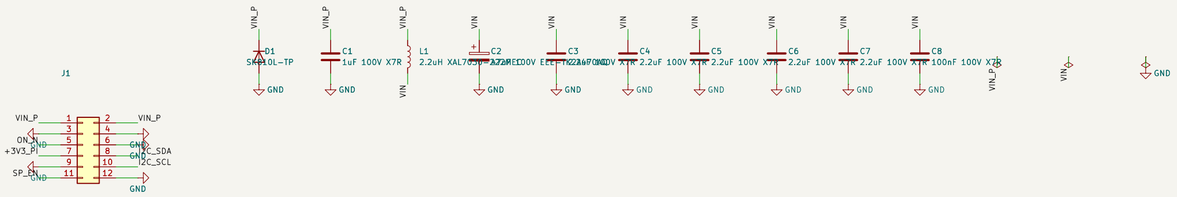
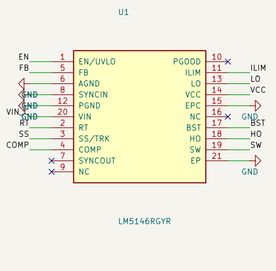
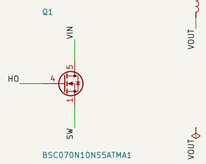
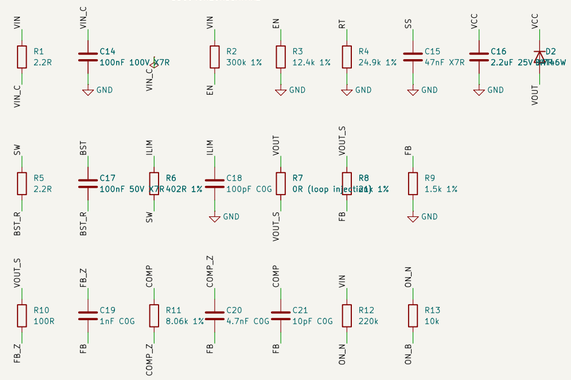
12 V buck: LM5146, 48 V to 12 V, 12 A, 400 kHz







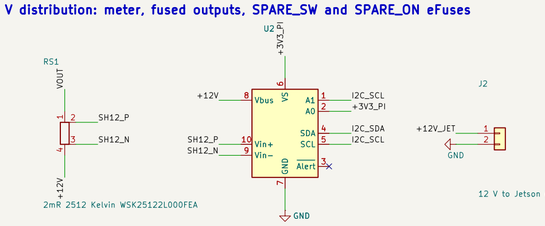


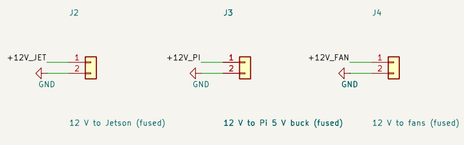


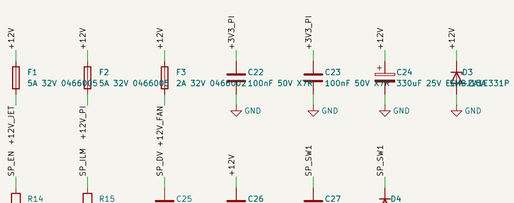


Input: SK810L crowbar (a reversed pigtail clamps at about 1 V; U7 on the main board limits and times out), 1 uF at the connector, 2.2 uH filter, 47 uF aluminium (0.42 Ohm ESR damps the filter and the pigtail), 5 x 2.2 uF hot-loop ceramics at Q1. tools/absorber_sim.py (main board) runs the bank-trip spike through U7, F2, the pigtail and this filter.
UVLO: 300k / 12.4k: on at 30.2 V, off at 27.2 V (10 uA hysteresis current). Below the 30.5 V worst sag at 90 A from a 38 V pack, so motor load can't brown out compute before the BMS does; the Pi still shuts down on its load-aware threshold.
ON/OFF: the main board's 2N7002 pulls pin 5 low through 220R when U7's PGD is good. Pin 5 open: 220k from VIN biases the NPN, which holds EN low (off). Pin 5 low: NPN off, EN rises: on. Pin 5 idles at about 2.7 V, like the DVE's.
Power train (TI design 2 uprated): 400 kHz (RT 24.9k), 6.8 uH 8.9 mOhm 21.8 A Isat, 3.3 A ripple; Q1 BSC070N10NS5 (low Qgd), Q2 BSC040N10NS5. Valley limit by Q2 RDS(on): 200 uA x 402R / 4.5 mOhm = 18 A cold, about 12–13 A hot: over the 11.6 A ceiling, under the 19–21 A of the inductor. Hiccup on overload.
Output: 12.0 V (0.8 V x (1 + 21k / 1.5k)); 5 x 22 uF here, plus 330 uF + 2 x 22 uF on the main board at the far end of the pigtail. Compensation is TI's for this L and C (40 kHz crossover). The pigtail and 330 uF resonate near 20 kHz: measure the loop with an injection resistor in place of the 0R before trusting it. VCC from VOUT through BAT46W (TI's D1) cuts bias loss.
Loss at 12 A, 48 V (est): Q1 0.4 W conduction + 1.2 W switching, Q2 0.7 W, inductor 1.3 W, controller 0.3 W: about 4 W, 97 %. Board: about 60 x 85 mm with the 12 V distribution and its six outputs, bolted to the rail with a thermal pad like the DVE.

12 V distribution: meter, fused outputs, SPARE_SW and SPARE_ON eFuses







Replaces the DVE0125V2NBC on J24. Main board: hardware/power-board/schematic.

marengo

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Title: marengo 12 V buck board (rev B5)

Size: A2	Date: 2026-11-06	**Rev: B-wip**
KiCad E.D.A. 10.0.8		Id: 1/1